

Johnny García Betancourt

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MSc student specialized in system modeling and control with a deep passion for differential geometry and theoretical physics applied to robotics. My work focuses on developing energy-based control frameworks that prioritize passivity and intrinsic task safety through a physics-informed approach. Currently, my research bridges these rigorous control strategies with machine learning and image processing for robust fault diagnosis. I am highly motivated to apply my mathematical and mechatronic expertise to solve the high-precision challenges inherent in medical robotics.

Research Interests

System Modeling & Control, Geometric Mechanics, Theoretical Physics applied to Robotics, Energy-Based Control (Passivity & Intrinsic Safety), Fault Diagnosis (ML & Image Processing), and Mechatronics.

Education

- Master of Science in Electrical Engineering (Control & Robotics)** 2025 – Present
King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia GPA: 3.96/4.00
- **Thesis:** Bridging Physical Safety and Active Tolerance in Robotic Manipulators: A Hierarchical Energy-Aware Architecture.
 - **Relevant Coursework:** Geometric Control for Robotics, Optimal Control, Linear Control Systems, Classical Mechanics, Adv. Methods for Theoretical Physics, Image Processing, Robot Perception.
- Master in Automation and Control** 2024
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador GPA: 99.46/100
- **Thesis:** Self-Diagnosable Automation and SCADA System for Industrial Mixing Tanks utilizing Predictive Maintenance and Sensor-based Health Monitoring.
 - **Relevant Coursework:** Machine Learning-Based Control, Identification & Control of Dynamic Systems, Industrial Robotics & Computer Vision, Embedded Control Systems.
- B.S. in Mechatronics Engineering** 2017 – 2022
ESPOL, Ecuador
- **Thesis:** Mechanical and Electrical Design, and Automation of an Industrial Liquid Filling Machine.

Research Projects

- Bridging Physical Safety and Active Tolerance in Robotic Manipulators: A Hierarchical Energy-Aware Architecture** Thesis Project | Currently Working
- **Focus:** Development of a hierarchical framework that enforces deterministic physical safety via Port-Hamiltonian Energy Tanks while maintaining active fault tolerance through differential flatness-based re-planning on a 6-DOF Kinova Gen3.
- Spectral Signatures of Robot Faults: A Port-Hamiltonian Visual Diagnosis Framework** Currently Working
- **Focus:** Transformation of 1D Hamiltonian power residuals into 2D Time-Frequency Scalograms via Continuous Wavelet Transform (CWT) for CNN-based fault isolation of LoE, friction, and vibrations.
- Fault-Tolerant Flatness-Based MPC via Dynamic Torque-to-Acceleration Constraint Mapping** Under Review
- **Focus:** Exploiting differential flatness to obtain canonical linear prediction models and implementing dynamic constraint-mapping from physical torque limits to state-dependent acceleration bounds.

Academic Experience

Electronics Laboratory Assistant

2025 – Present

King Fahd University of Petroleum and Minerals (KFUPM)

- Guided undergraduate students through hands-on electronics laboratory experiments, providing practical instruction on circuit analysis and instrumentation.

Statistics Teaching Assistant

2022

ESPOL, Facultad de Ingeniería Mecánica y Ciencias de la Producción (FIMCP)

- Instructed engineering students on statistical analysis and developed academic material for curriculum support.

Professional Experience

Detecuator S.A.

2022 – 2024

Maintenance Coordinator

- Managed mechatronic projects and utilized 2D/3D CAD for industrial infrastructure and automation solutions.

SEOR S.A.

2023 – 2024

Assistant Mechanical Designer

- Focused on the design and management of engineering plans and automation system integration.

Technical Skills

Control & Modeling: High-Fidelity System Modeling, Geometric Control, Differential Geometry, Port-Hamiltonian Systems, Differential Flatness, Optimal Control, Passivity Based Control.

Perception & Estimation: Invariant Kalman Filter on Manifolds, Pose Optimization, Fundamentals of SLAM.

Mechatronics & Design: Mechatronic System Design, Advanced Industrial Instrumentation, Hardware Integration, Actuator Dynamics.

Software & Tools: MATLAB/Simulink, Python, SolidWorks, ANSYS, LaTeX.